

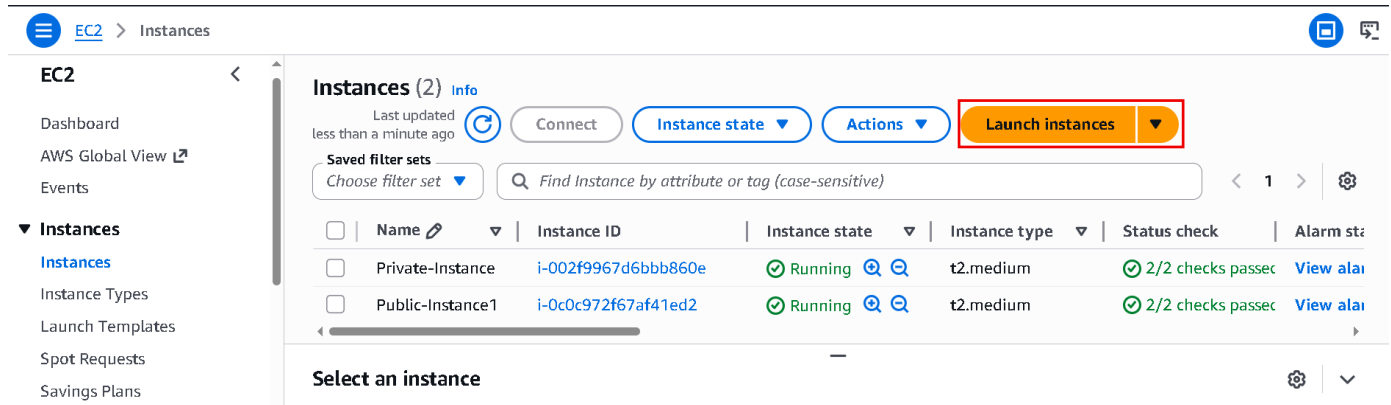
EBS Snapshot Backup: -

Summary: -

In this lab, I created a Windows EC2 instance and added extra storage using an EBS volume. I stored files in the new drive and created a snapshot as a backup. After deleting the files, I restored the backup by creating a new volume from the snapshot and attached it back to the instance.

Let's Create: -

- Search EC2 in Aws Console and Open it.
- Now Let's create windows EC2 instance.



EC2 > Instances

Instances (2) Info

Last updated less than a minute ago

Connect Instance state Actions Launch instances

Saved filter sets Choose filter set

Find Instance by attribute or tag (case-sensitive)

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐	Private-Instance	i-002f9967d6bbb860e	Running	t2.medium	2/2 checks passed	View alarm
☐	Public-Instance1	i-0c0c972f67af41ed2	Running	t2.medium	2/2 checks passed	View alarm

Select an instance

- Now opens EC2 in new tab.
- And click Launch instance.

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

Add additional tags

- Enter instance name.
- Scroll down.

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

 Search our full catalog including 1000s of application and OS images

Quick Start


Amazon Linux



Mac


ubuntu


Microsoft


Red Hat


SUSE


[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

- Choose OS click Browse more AMIs.

☰ [EC2](#) > [Instances](#) > [Launch an instance](#) > AMIs ⓘ

 Search for an AMI by entering a search term e.g. "Windows" 

Quick Start AMIs (45)

Commonly used AMIs

My AMIs (13)

Created by me

AWS Marketplace AMIs (6091)

AWS & trusted third-party AMIs

Community AMIs (500)

Published by anyone



Windows

Free tier eligible

Verified provider

Microsoft Windows Server 2019 Base

ami-08e30c01541e6a4f3 (64-bit (x86))

Microsoft Windows 2019 Datacenter edition.
[English]

Platform: windows
Virtualization: hvm

Root device type: ebs
ENA enabled: Yes

Select

 64-bit (x86)

- I am going to use windows 2019 Base.
- Click on select and check first they are Free tier eligible or not.

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

 Search our full catalog including 1000s of application and OS images

AMI from catalog

Recents

My AMIs

Quick Start

Name

Microsoft Windows Server 2019 Base

Verified provider

Free tier eligible



[Browse more AMIs](#)

Including AMIs from
AWS, Marketplace and
the Community

Description

Microsoft Windows 2019 Datacenter edition. [English]

Microsoft Windows Server 2019 with Desktop Experience Locale English AMI provided by Amazon

Image ID

ami-08e30c01541e6a4f3

- Now scroll down.

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand SUSE base pricing: 0.0104 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour

On-Demand RHEL base pricing: 0.0392 USD per Hour

On-Demand Windows base pricing: 0.0196 USD per Hour

On-Demand Linux base pricing: 0.0104 USD per Hour

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

- now we are just going to select this t3.micro
- In this we are going to get two virtual CPU and one GB Ram.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Select

 [Create new key pair](#)

- Click Create new key pair.

Create key pair

Key pair name

Key pairs allow you to connect to your instance securely.

demo

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

Cancel

Create key pair

- Enter key name.
- Key pair type as RSA
- And .pem file
- And last click Create key pair
- Now your keypair is downloaded and save it.
- Now scroll down.

▼ Network settings

Info

Edit

Network

Info

vpc-072c4acd761c3b942

Subnet

Info

No preference (Default subnet in any availability zone)

Auto-assign public IP

Info

Enable

Additional charges apply when outside of free tier allowance

- Now click Edit.

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-072c4acd761c3b942
172.31.0.0/16

(default) ▼



Subnet [Info](#)

subnet-04ffa8e5e9c79ad3d

VPC: vpc-072c4acd761c3b942 Owner: 463646775279

Availability Zone: ap-south-1a (aps1-az1) Zone type: Availability Zone

IP addresses available: 4089 CIDR: 172.31.32.0/20



[Create new subnet](#)

Auto-assign public IP [Info](#)

Enable ▼

[Additional charges apply](#) when outside of [free tier allowance](#)

- Now Default VPC not change it.
- Select Subnet ab-south-1a because in Mumbai region three availability zones 1a, 1b, 1c if we not choose then aws decide in which availability zones they want to create our EC2 instance.

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - required

launch-wizard-80

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and . _ - / () # , @ [] + = & ; ! \$ *

Description - required [Info](#)

launch-wizard-80 created 2026-05-19T08:26:27.229Z

- Now here create a security group or you select your created security group also.
- Aws always create a custom security group if you want to change name of security group you can change it.

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 3389, 0.0.0.0/0)

[Remove](#)

Type [Info](#)

rdp ▼

Protocol [Info](#)

TCP

Port range [Info](#)

3389

Source type [Info](#)

Anywhere ▼

Source [Info](#)

Add CIDR, prefix list or securi

0.0.0.0/0

Description - optional [Info](#)

e.g. SSH for admin desktop

- You can add extra permissions in your security group.
- By default for windows rdb permissions is added.
- Now scroll down.

▼ Configure storage [Info](#)

Advanced

1x GiB ▼ Root volume, 3000 IOPS, Not encrypted

1x GiB ▼ EBS volume, 3000 IOPS, Not encrypted

[Remove](#)

[i](#) Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage [X](#)

[Add new volume](#)

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

- Suppose if you are going to buy a laptop, they will ask you right that what is the storage Requirement like One TB or 512 GB etc.
- Same way here.
- It will provide you 30 GB volume.
- Now this volume is actually root volume.
- It means it will install operating system inside this volume.
- We can add new volume from here also by clicking ADD new volume.
- I added 15gb extra.
- Now simply scroll down.

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 30 GiB

[Cancel](#)

[Launch instance](#)

[Preview code](#)

- And click Launch instance.

Instances (1) [Info](#)

Last updated
less than a minute ago



[Connect](#)

[Instance state](#) ▼

[Actions](#) ▼

[Launch instances](#) ▼

Saved filter sets

[Choose filter set](#) ▼

[Find Instance by attribute or tag \(case-sensitive\)](#)

< 1 >



☐	Name ✎ ▼	Instance ID	Instance state ▼	Instance type ▼	Status check	Availabili
☐	WindowsInsta...	i-0790663b2d2e53b9b	Running 🔍 🔍	t3.micro	Initializing	ap-south-

- Now you see our instance is created but current status of our instance is initializing so wait for status check become 3/3 checks passed

Instances (1/1) [Info](#)

Last updated less than a minute ago  [Connect](#) [Instance state ▼](#) [Actions ▼](#) [Launch instances ▼](#)

Saved filter sets
Choose filter set ▼

Find Instance by attribute or tag (case-sensitive)

☒	Name 	Instance ID	Instance state	Instance type	Status check	Availability
☒	WindowsInsta...	i-0790663b2d2e53b9b	Running  	t3.micro	 3/3 checks passed	ap-south-

i-0790663b2d2e53b9b (WindowsInstance)



Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance summary [Info](#)

Instance ID

 i-0790663b2d2e53b9b

Public IPv4 address

 13.232.250.79 | [open address](#) 

Private IPv4 addresses

 172.31.45.220

- Now see our status check becomes 3/3 checks passed.
- Now connect with our Machine.
- Select the created instance click on connect.

Connect [Info](#)

Connect to an instance using the browser-based client.

Instance ID  i-0790663b2d2e53b9b (WindowsInstance)	VPC ID  vpc-072c4acd761c3b942	Security groups  sg-06110383e378abc30 (launch-wizard-277)	IAM role —
--	--	--	----------------------

SSM Session Manager [RDP client](#) EC2 serial console

- Now click RDP client.
- And scroll down.

Instance ID

 i-0790663b2d2e53b9b (WindowsInstance)

Connection Type

☒ Connect using RDP client

Download a file to use with your RDP client and retrieve your password.

☐ Connect using Fleet Manager

To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed and running on the instance. For more information, see [Working with SSM Agent](#) 

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

[Download remote desktop file](#)

When prompted, connect to your instance using the following username and password:

Public DNS

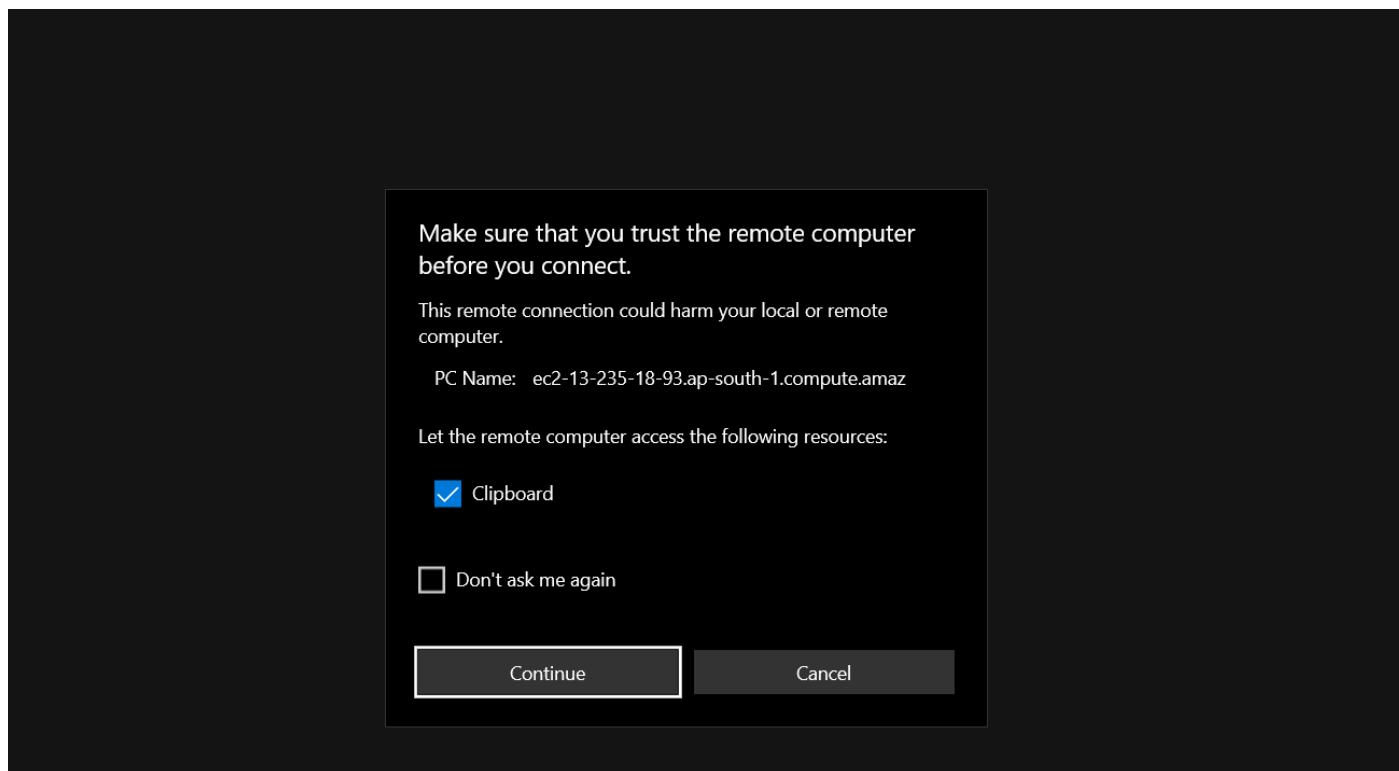
 ec2-13-232-250-79.ap-south-1.compute.amazonaws.com

Username [Info](#)

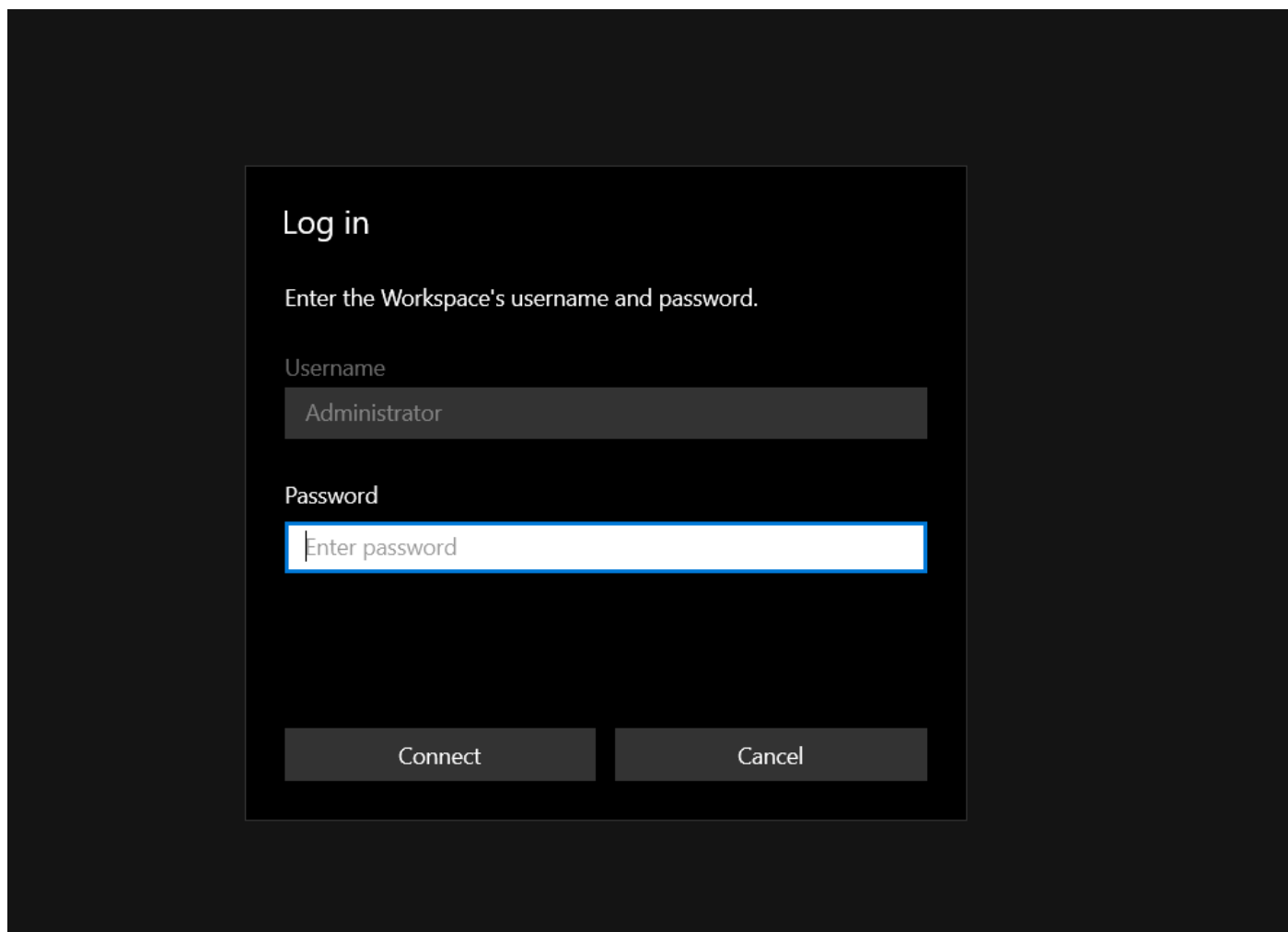
 Administrator ▼

Password [Get password](#)

- Now click on download remote desktop file.
- When your file is downloaded go to download folder and open it.



- Now click continue.



- Now we need a password.
- Remember you created a demo.pem key in starting now we use it.
- Go to your browser instance tab.

Instance ID

 i-0790663b2d2e53b9b (WindowsInstance)

Connection Type

- ☒ Connect using RDP client
Download a file to use with your RDP client and retrieve your password.

- ☐ Connect using Fleet Manager
To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed and running on the instance. For more information, see [Working with SSM Agent](#) 

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

 [Download remote desktop file](#)

When prompted, connect to your instance using the following username and password:

Public DNS

 ec2-13-232-250-79.ap-south-1.compute.amazonaws.com

Username [Info](#)

 Administrator 

Password [Get password](#)

- Now click on get password.

Get Windows password [Info](#)

Use your private key to retrieve and decrypt the initial Windows administrator password for this instance.

Instance ID

 i-0790663b2d2e53b9b (WindowsInstance)

Key pair associated with this instance

 demo

Private key

Either upload your private key file or copy and paste its contents into the field below.

 [Upload private key file](#)

Private key contents

- Now click on upload private key file.
- So you upload your demo.pem file.

 [Upload private key file](#)

demo.pem
1.67 KB 

Private key contents

```
-----BEGIN RSA PRIVATE KEY-----
MIIEowIBAAKCAQEAyv236ltS1zMgilPBjRBrK5qInIJ5LXDmX9bdpc6O+jrL5+t2
HcGPhXN/Doy3yaFD2Tm5N8OUpcNv6vh4LWBP7RBXe59bQgLepILZ+BsB+fQ02oAw
kfeIEBxHC+5Rpz8Dm92rsOJxhfw/8+75K1wD4ywRhrtSRKcqZu4LxsWmE2HCoEhZ
eWn8yE/mmgCUIv1XjsftthocEBvcy1RK63RB+F58SyyXa7XudRzqZM46k9xncurFv
MxKfZHT+e/diwVDrAH/Ggt2+yep4U5LxqUK2WrusG0PnkOFFxAn8P6igS+Jr5ru
KS7S0veZ/pROglhmhLO2qUL46VHXzMnGjFlgQwIDAQABAoIBAGFvWpZKxLZYtO8c
z49B0wr5qVv/DAbso/qcyTK/cB2kRUYpz3Zauvdf9p/blh9dEBoP4rfaJULzrPL4
```

[Cancel](#)

[Decrypt password](#)

- Now simply scroll and click Decrypt password.

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

[Download remote desktop file](#)

When prompted, connect to your instance using the following username and password:

Public DNS

 ec2-13-235-18-93.ap-south-1.compute.amazonaws.com

Username [info](#)

 Administrator

Password

 @JNfy?%;z=wJhuaaavg;!%oukFy;gi8Dv

 If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.

- Now simply copy the Password.

Log in

The password used to connect to the remote resource didn't work.

Enter a different password and try again.

Username

Administrator

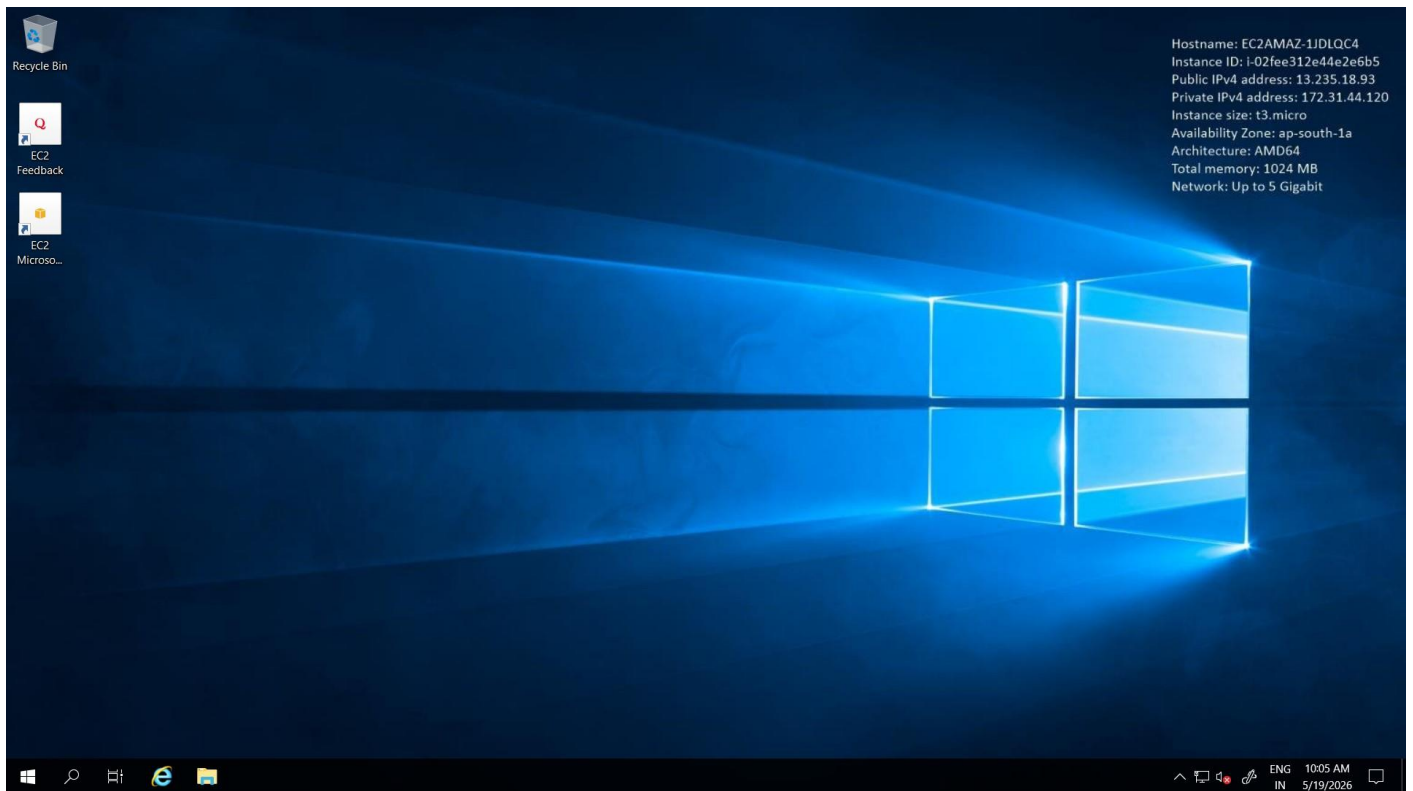
Password

.....

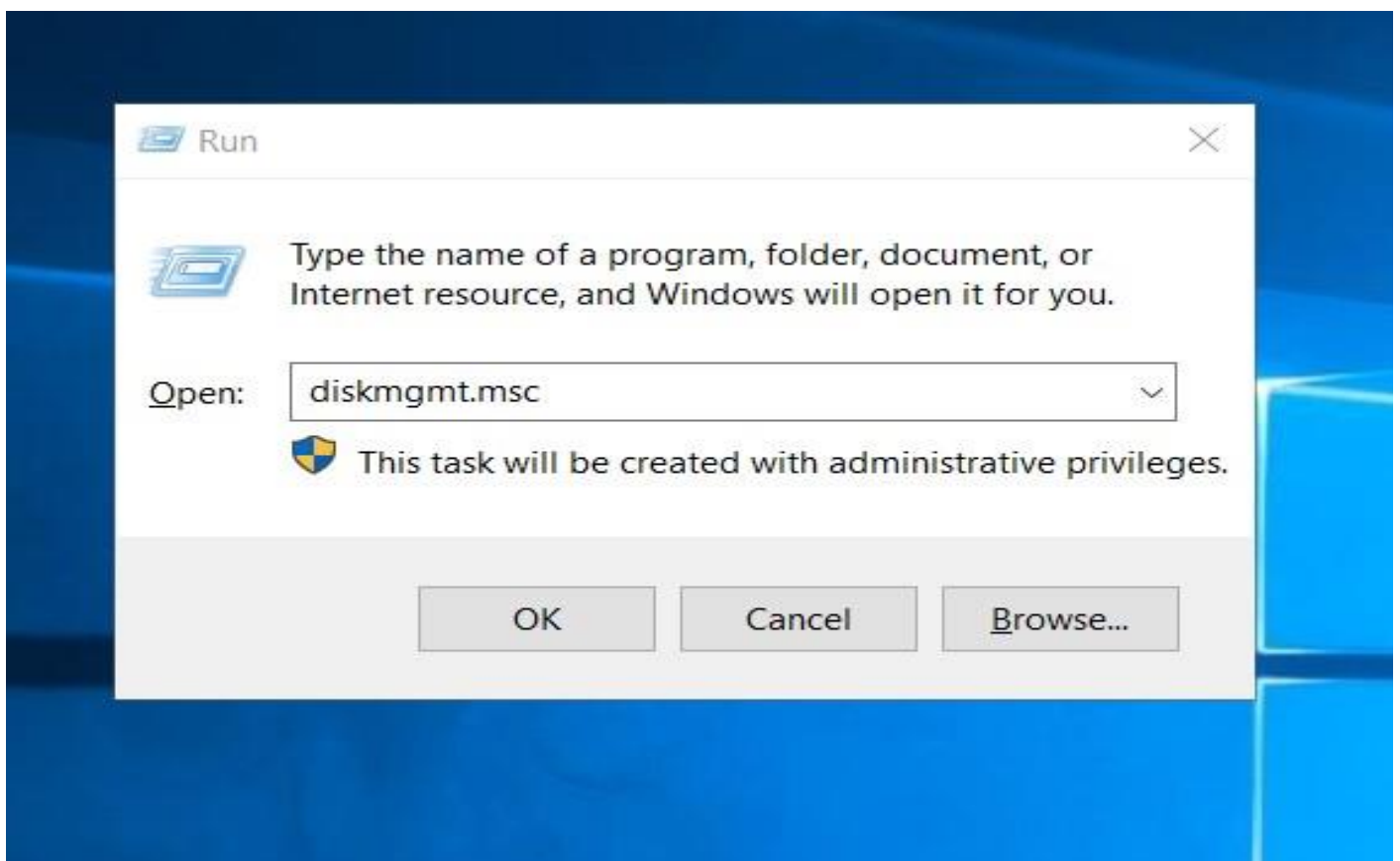
Connect

Cancel

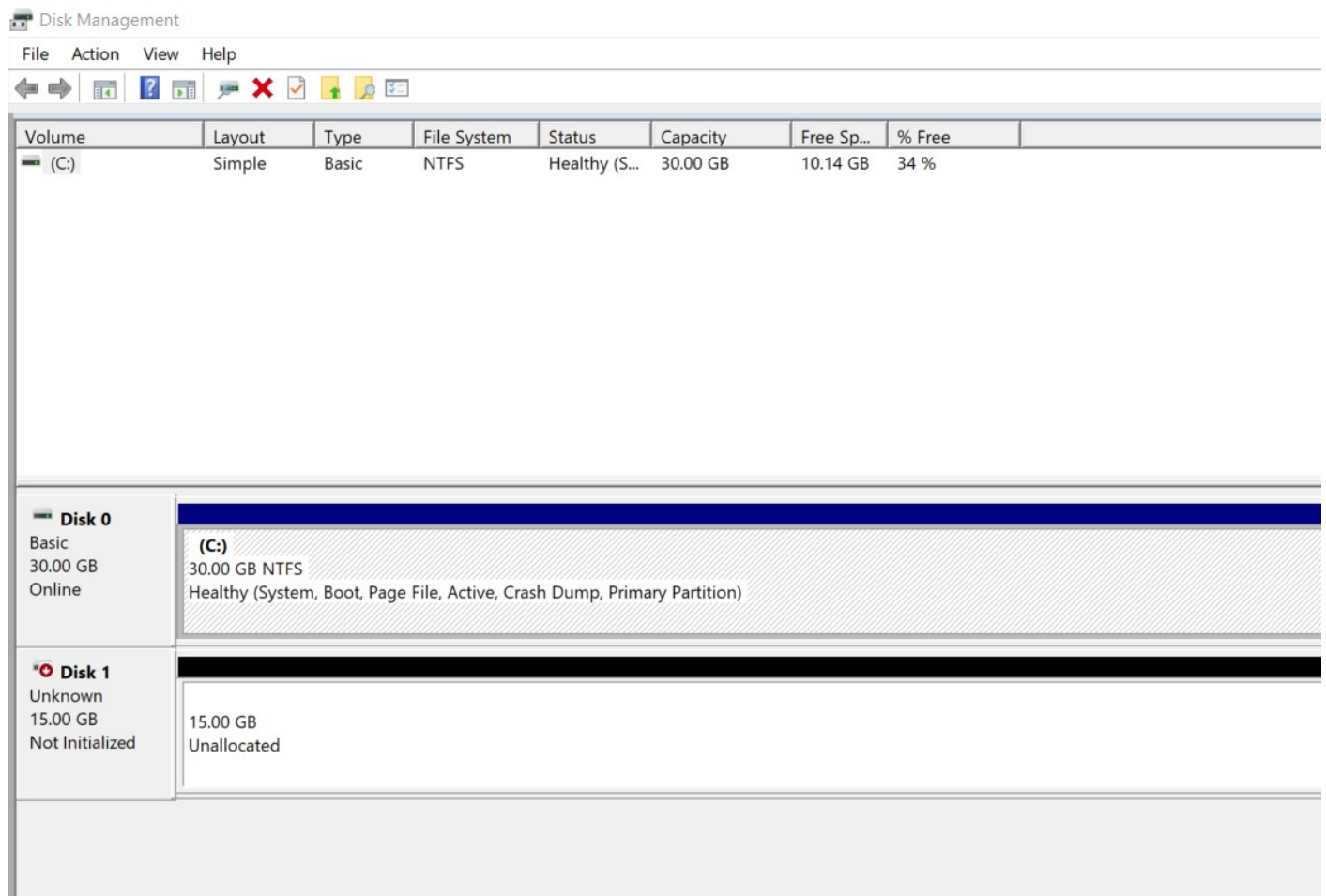
- Now paste the password in remote Desktop and click connect.



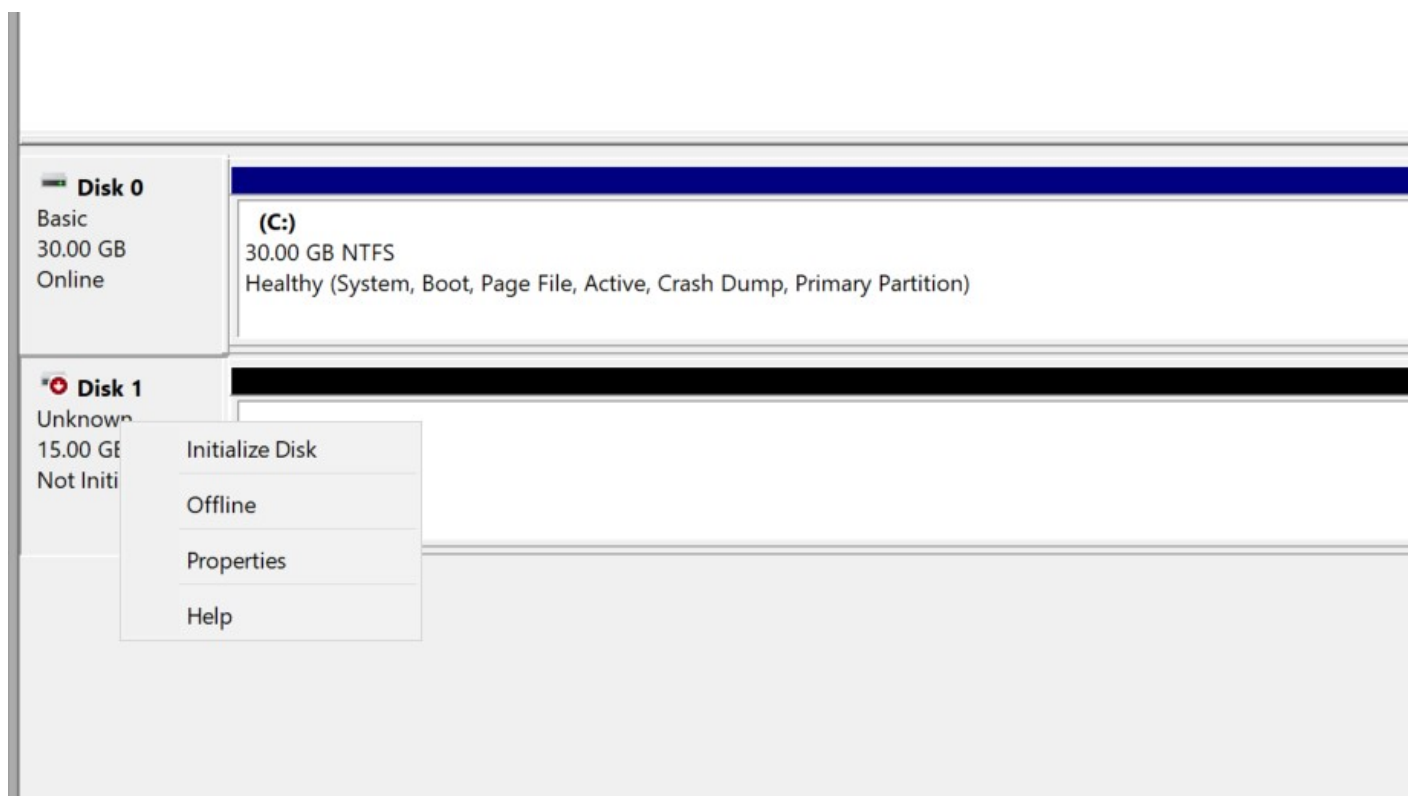
- Now you see this is the window machine currently we created.
- Now press window key + r



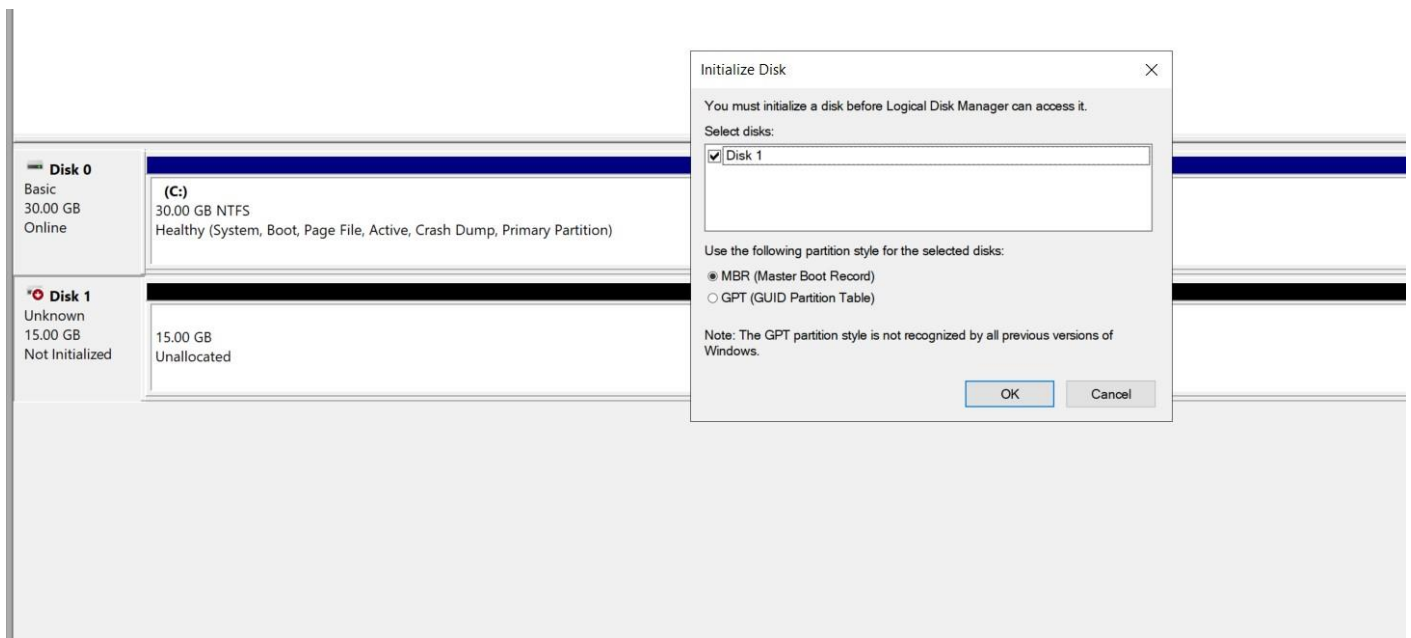
- Then type diskmgmt.msc in run terminal.



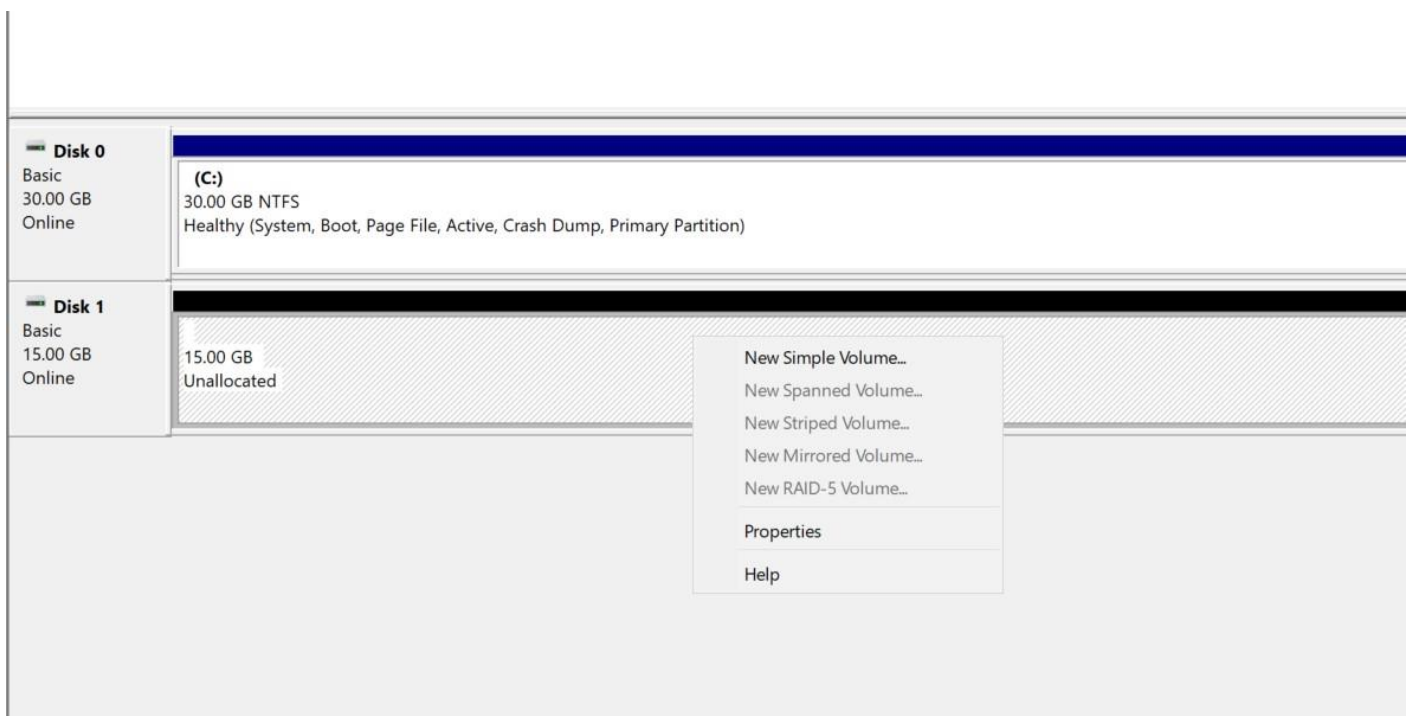
- Now this is added but you go to This Pc its's not visible.



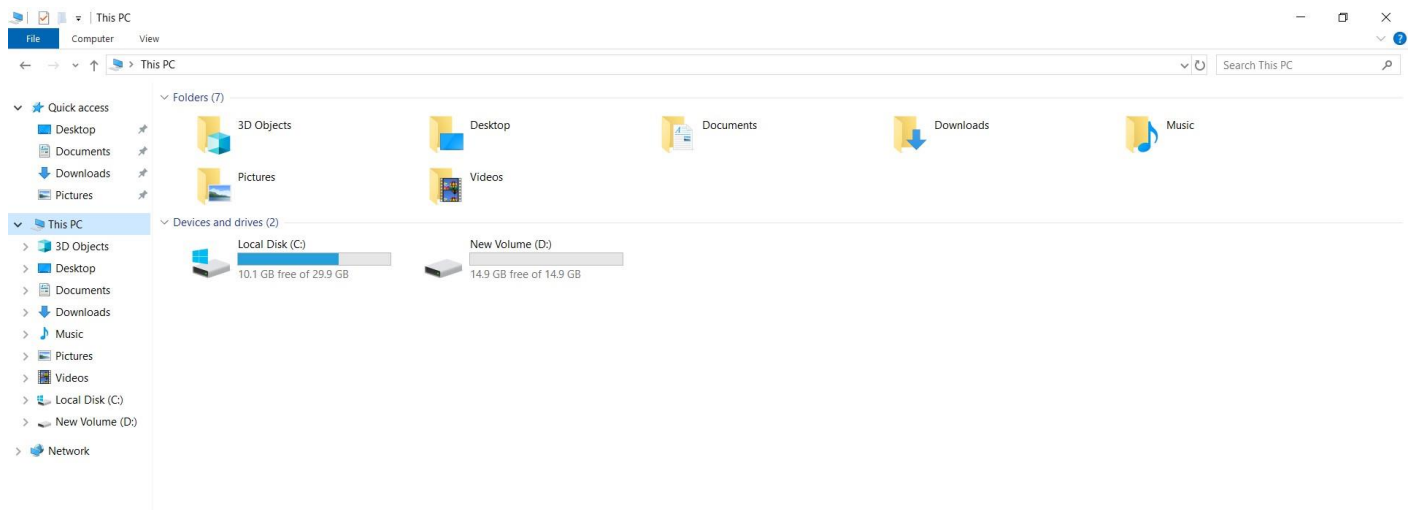
- For that this driver shows in my pc right click on it.
- Then select initialize Disk.



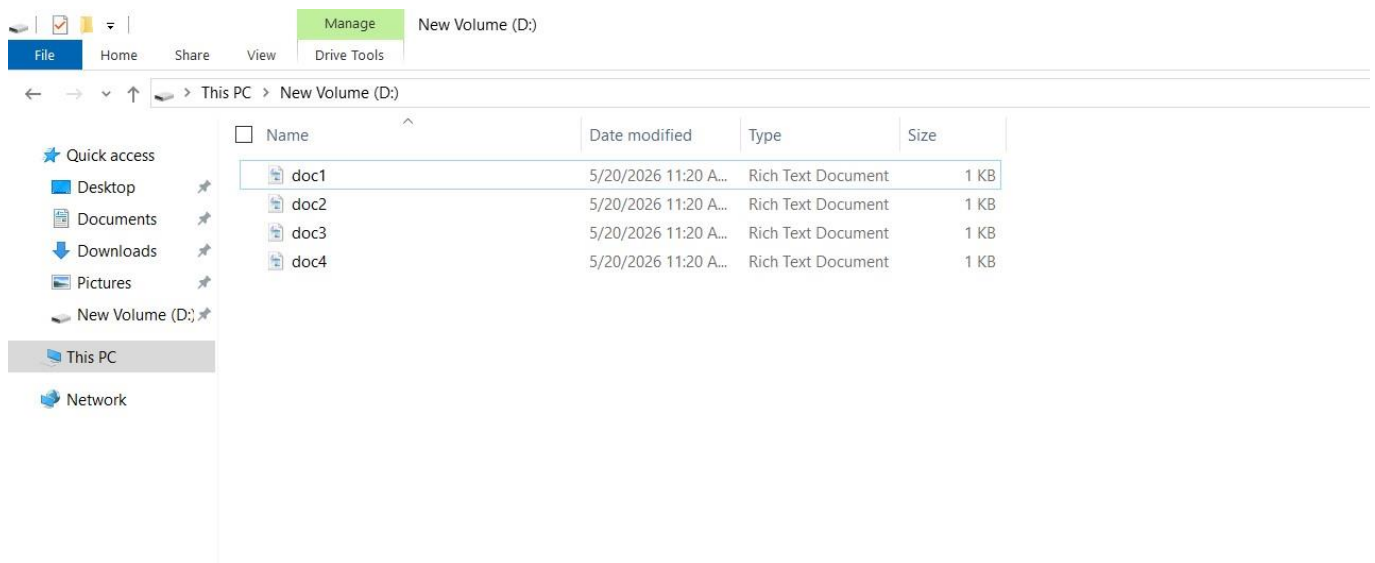
- Now simply click ok.



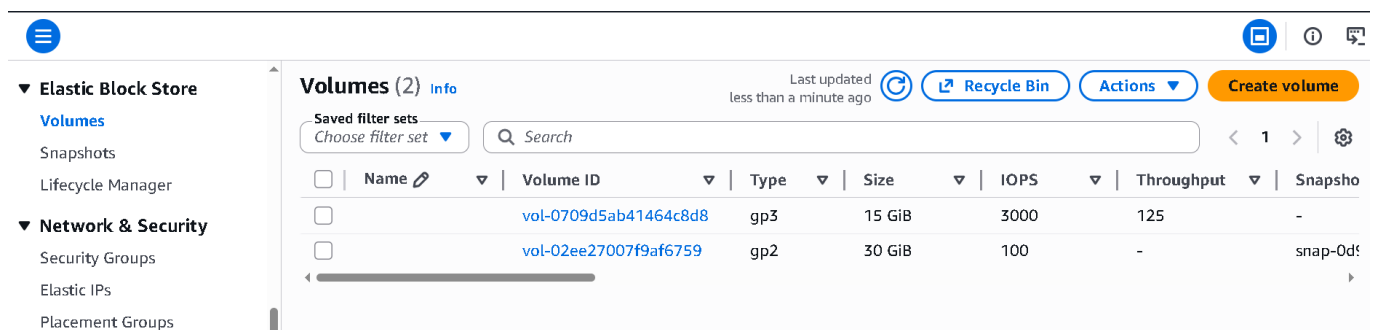
- Now right click on unallocated side and select New Simple Volume.
- Then click next. Next and last finish.
- All process is done now our new drive shows in this pc.
- So go inside This Pc.



- Now you see your new drive is visible in this pc.



- Now create some Rich Text documents and save in new storage.
- Now I am worried about my data and I want to take a backup of this.
- Open your real window and open the Ec2 tab.



- Now click Volumes inside Elastic Block Store in nav bar.
- Click name on 15gb and give name (MyBackupVolume).
- Then after giving name click Snapshot in nav bar.

▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

▼ Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Snapshots (4) Info

Last updated
less than a minute ago

Recycle Bin

Actions

Create snapshot

Snapshot scope

Owned by me

Search

< 1 > ⚙

☐	Name	Snapshot ID	Full snapshot size	Volume size	Description
☐		snap-04c566df00e81bd05	29.67 GiB	30 GiB	Created by CreateImage(i-...
☐	mynsnapshot1	snap-09c7e9adab0ca0651	21 MiB	5 GiB	for demo
☐		snap-0e2299432e55e03b2	33.5 MiB	10 GiB	for demo
☐		snap-078e6c31fb8d33fad	1.7 GiB	8 GiB	Created by CreateImage(i-...

- Click Create Snapshot.

Create snapshot Info

Create a point-in-time snapshot of an EBS volume and use it as a baseline for new volumes or for data backup. You can create snapshots from an individual volume, or you can create multi-volume snapshots from all of the volumes attached to an instance.

Source

Resource type Info

☒ Volume

Create a snapshot from a specific volume.

☐ Instance

Create multi-volume snapshots from an instance.

Volume ID

The volume from which to create the snapshot.

vol-0709d5ab41464c8d8 (MyBackupVolume)
aps1-az1 (ap-south-1a)



- Now Select Volume ID (MyBackupVolume)
- Scroll down.

Snapshot details

Description

Add a description for your snapshot.

learning snapshot

255 characters maximum

Encryption Info

Not encrypted

- Give description.

Tags Info

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add up to 50 tags.

Cancel

Create snapshot

- Now simply scroll down and click Create snapshot.

Snapshots (5) Info

Last updated
1 minute ago



Recycle Bin

Actions

Create snapshot

Snapshot scope

Owned by me

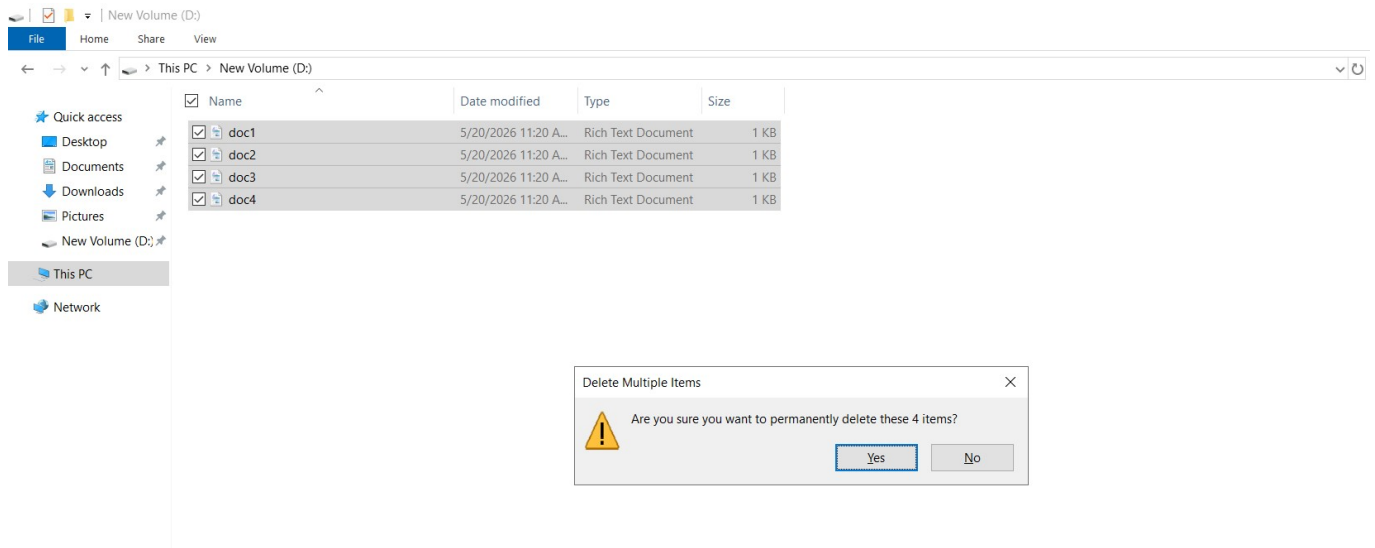
Search

< 1 >



☐	Name	Snapshot ID	Full snapshot size	Volume size	Description
☐	MyBackup	snap-0bd28c5764fe2a89a	43 MiB	15 GiB	learning snapshot

- It's created I also give a name.
- Now go to your Machine.



- And delete all documents.
- Now go back to real window and open browser and go inside snapshot tab.

Snapshots (1/5) Info

Last updated
6 minutes ago



Recycle Bin

Actions

Create snapshot

Snapshot scope

Owned by me

Search

< 1 >



☒	Name	Snapshot ID	Full snapshot size	Volume size	Description
☒	MyBackup	snap-0bd28c5764fe2a89a	43 MiB		
☐		snap-04c566df00e81bd05	29.67 GiB		
☐		snap-09c7e9adab0ca0651	21 MiB		
☐		snap-0e2299432e55e03b2	33.5 MiB		
☐		snap-078e6c31fb8d33fad	1.7 GiB		

Create volume from snapshot

Create image from snapshot

Copy snapshot

Launch copy duration calculator

Delete snapshot

Manage tags

Snapshot settings

Archiving

- Select your snapshot and click Actions and select Create volume from snapshot.

Snapshot summary Info



Click refresh to view backup information

The volume type that you select and the tags that you assign determine whether the volume will be backed up by any Data Lifecycle Manager policies.

Cancel

Create volume

- Now simply scroll down and click create volume.
- And go to volume section.

Volumes (1/3) Info Last updated 1 minute ago [Recycle Bin](#) [Actions](#) [Create volume](#)

Saved filter sets [Choose filter set](#)

	Name	Volume ID	Type	Size
☐	MyBackupVol...	vol-0709d5ab41464c8d8	gp3	15 GiB
☐		vol-02ee27007f9af6759	gp2	30 GiB
☒		vol-0e603c014c6d9712d	gp3	15 GiB

Volume ID: vol-0e603c014c6d9712d

[Details](#) [Status checks](#) [Monitoring](#) [Tags](#)

Volume ID	Size	Type
vol-0e603c014c6d9712d	15 GiB	gp3

Actions menu:

- Modify volume
- Create snapshot
- Create snapshot lifecycle policy
- Delete volume
- Attach volume**
- Detach volume
- Force detach volume
- Manage auto-enabled I/O
- Copy volume - new
- Manage tags
- Resilience testing**
 - Previously fault injection

- Now we have three volumes, but new volume is not attached.
- So, select new volume and click Actions and select Attach volume.

Basic details

Volume ID
 vol-0e603c014c6d9712d

Availability Zone
aps1-az1 (ap-south-1a)

Instance [Info](#)
i-0a6d925fdbae018db (WindowsInstance) (running)

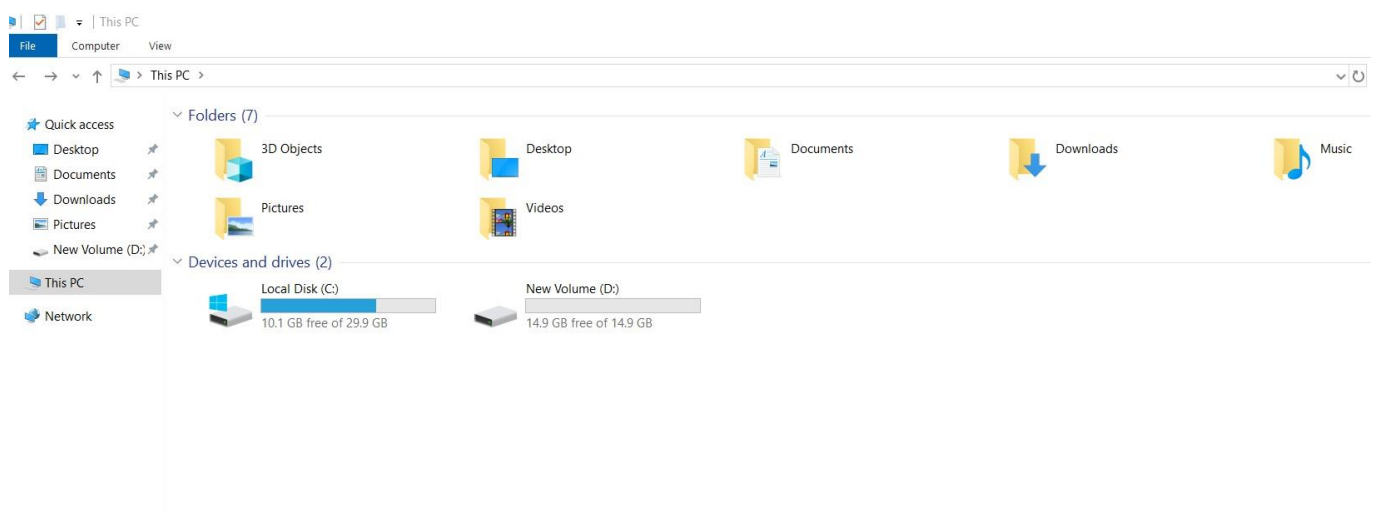
Only instances in the same Availability Zone as the selected volume are displayed.

Device name [Info](#)
xvdf

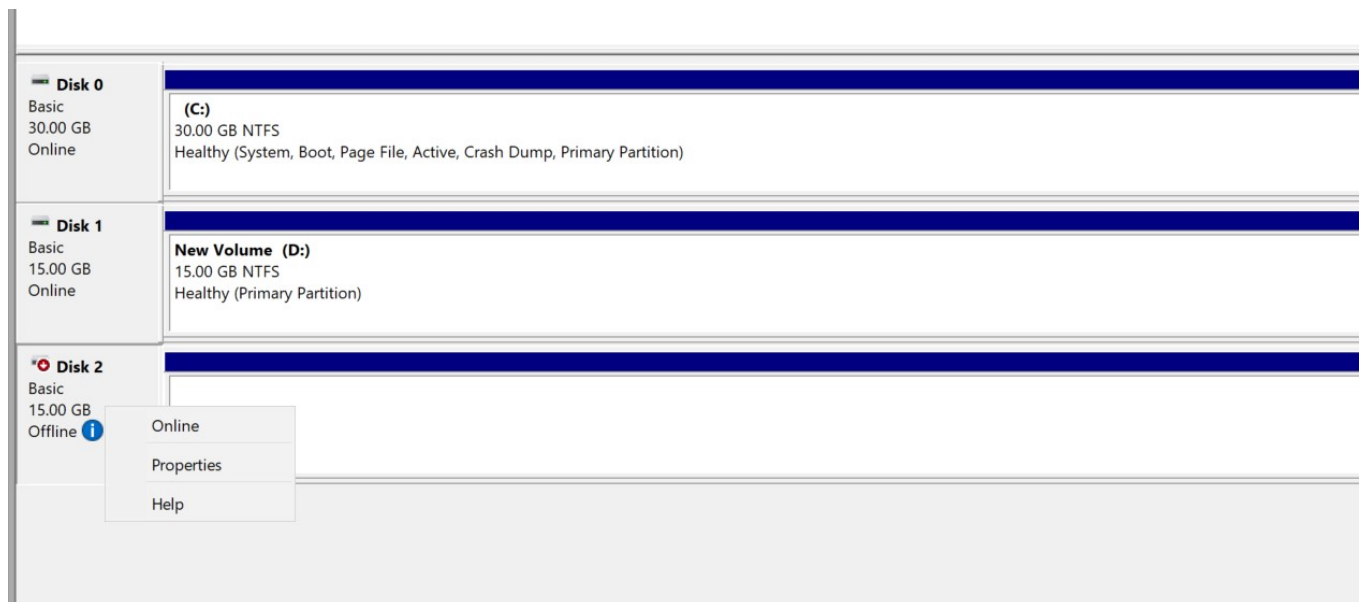
Recommended device names for Windows: /dev/sda1 for root volume. xvdf[f-p] for data volumes.

[Cancel](#) [Attach volume](#)

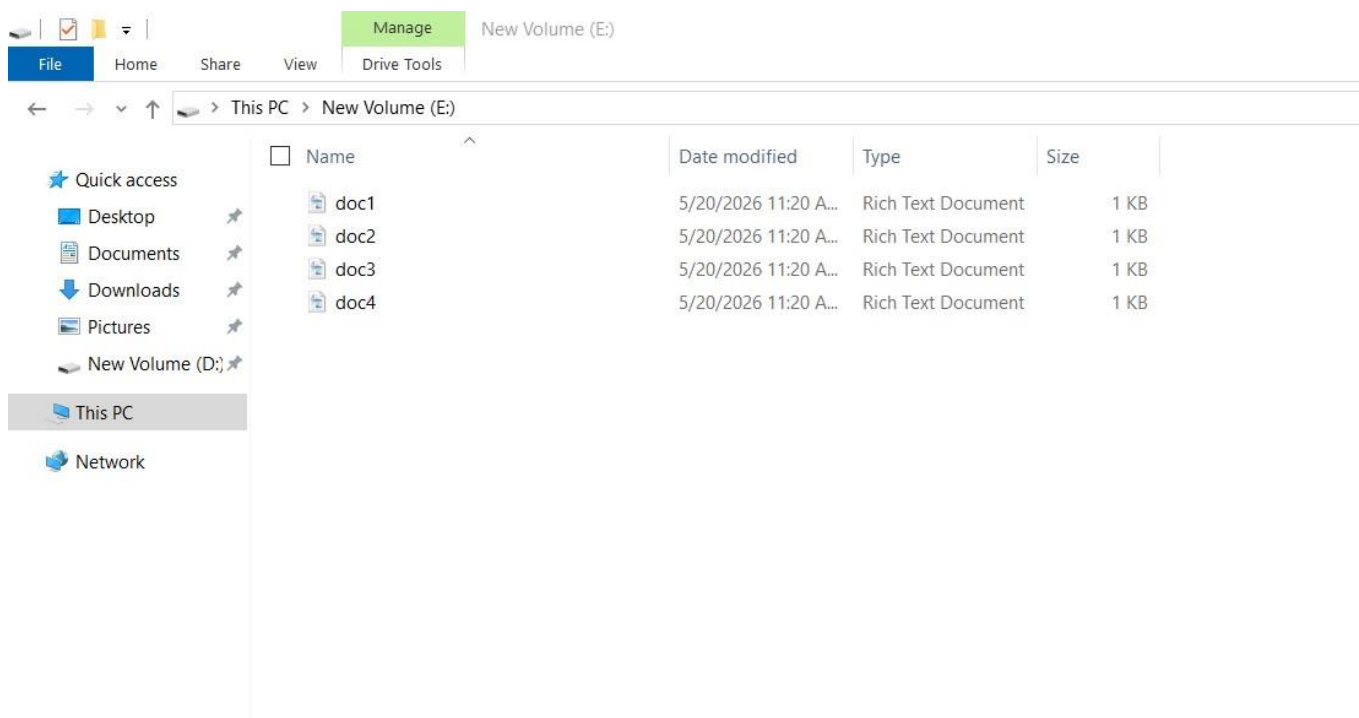
- Now select your instance and click Attach volume.
- Now go to virtual machine.



- Now currently showing two driver we need to open disk management.



- Now right click on disk 2.
- And select Online.
- Now go back to this pc and open new drive.



- Now see we backup our documents.